



Play with Fire

Mónica Mendes
FBA, University of Lisbon
and CIEBA
Lisbon, Portugal
monica.mendes@fba.ul.pt

Pedro Ângelo
Audiência Zero
Porto
Portugal
pangelo@void.io

Valentina Nisi
M-ITI, University of Madeira
Madeira
Portugal
valentina@uma.pt

Nuno Correia
FCT, New University
of Lisbon and IMG
Lisbon, Portugal
nmc@di.fct.unl.pt



Figure 1. Visual impression of Play with Fire artistic concept.

ABSTRACT

Play with Fire is an interactive art installation that proposes participants to ignite generative fires over live streaming video of selected forests. This experience paradoxically encourages playing with fire to stimulate awareness and prevention of fire related damages to the forests.

We envisage the installation triggering controversial feelings by combining the “beauty and danger” of a forest on fire. Our goal is to raise awareness for human causes in forest fires and effect attitude change towards environmental protection.

In this proposal we outline the concept and installation setup for the ACE 2011 Creative Showcase and Interactive Art track.

Categories and Subject Descriptors

H.5.0 [Information Interfaces and Presentation]: General.
H.5.1 [Information Interfaces and Presentation]: Multimedia.
Information Systems – Video.

General Terms

Design, Experimentation, Human Factors.

Keywords

Digital Art, Games for Change, Generative Graphics, Real-Time Video, Sustainability.

1. INTRODUCTION

Play with Fire is an interactive digital art installation conceived and designed in order to stimulate activism and raise awareness in the area of sustainability and climate change, particularly related to forest fire hazards. In order to foster awareness and stimulate activism, we decided to conceive and design *Play with Fire* as a digital art experience that happens over three different moments:

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an introduction to the installation and its theme with an invitation to interact with it; the performative active part where the participant engages with the gestural interface, and a reflective part taking the form of a mobile application, which will stay with the participant for a long time after the installation experience. With *Play with Fire* we are promoting interactions and experiences that invite the users not only to reflect on fire hazards in forest settings, but also to act in such a setting, in order to gain understanding and accountability for the generated results.

Play with Fire is part of *RTiVISS (Real-Time Video Interactive Systems for Sustainability)*, an exploratory research in digital media related to sustainability. Through this research, we are reflecting on several questions: how can we propose constructive approaches to the destructive dynamics that aggravate climate change? Can art foster awareness and respect for nature? [6]. Harnessing the impact and the potential of art and technology on society and the environment, we aim at creating a “forests showroom” experience through digital media, in order to inspire change in the lifestyle of the public. The outcomes of *RTiVISS* include an online platform, interactive installations and the design of an open source surveillance kit prototype.

2. RELATED WORK

Climate change and surveillance are very relevant and currently discussed topics in the areas of digital art and design. The role and potential of design research in the transition towards sustainability is being discussed in mainstream design and digital media art events, such as “Repair” (2010) and “Goodbye Privacy” (2007) at *Ars Electronica* [1], *Changing the Change* (2008) [2], and *Transmediale* (2009) putting the threat to the sustainability of our planet in perspective [9].

The *Games for Change* project hosts an archive with many good examples of the expressive power of the medium for environmental awareness [3]. The work of artists like La Molleindustria [4] that appropriate and subvert videogames as an interactive medium for persuasive purposes has also influenced our work.

Whereas real-time video has been mainly used as a functional tool for surveillance, for informational and safety purposes, the use of this resource has an enormous potential for artistic exploration. Some initial steps in this direction have been made by projects such as *Funky Forest*, an interactive ecosystem where children create trees with their body and divert the water to the trees to keep them alive [10]. *Parque* is also an interactive installation with an ecological message: “the growth of a forest is determined by the amount of attention it receives” [5]. *Hand from Above* [7] is an experience that playfully challenges our perception of spaces and objects, enabling virtual and real to coexist in real-time, demonstrating participants’ immediate engagement.

3. THE *PLAY WITH FIRE* EXPERIENCE

In *Play With Fire*, a surveillance camera will be setup at a selected forest location, transmitting real-time video to the installation space, a room inside of which the performative gestural interaction is contained.

One semi transparent wall is where the mash-up media and information about fire hazards and forests will be displayed. From the outside, the participants are seen as silhouettes of the arsonist – “fire wizards” performing inside the installation room forming a choreography with the changing media displayed. This is also functioning as an attractor for the audience to enter the interaction space. When the participant enters inside the installation room, the system asks him or her to place his/her mobile device on a special deck. Ready to face the screen, the audience member is presented with the start of a fire, triggered by positioning the phone over the deck, and he/she can get ready for the performative interaction.

The participant faces the main screen. A whole body gestural interface based on a depth camera captures the participant's movement. He/she goes straight into the interaction with the real-time video of the selected forest projected on the whole large scale wall. The sound of crackling fire attracts attention to the action of the fire and how it is spreading. As the user is prompted to act upon the virtual fire, it will react through real-time generative graphics and fire animation according to the gesture just performed. Intuitive and natural gestures for controlling fire (starting it, growing, concentrating, moving it, extinguishing) have been tested and selected in order to map the user engagement, still allowing further performative explorations.

After a certain amount of time, the fire takes over the trees and forest and develops its own behaviors. If at this point the audience does not intervene by performing gestures trying to extinguish the fire, the forest fire climaxes to a point of no return. Fire flames follow the trees structures. Fire has developed its own behavior and spread through the forest. The scene becomes a spectacle for the audience. Once the fire has climaxed, the amount of virtual damage to the forest will depend on how much the audience has played with the fire, how much it has watched the spectacle of burning trees go on and if he/she had ever tried to stop the flames from taking over. A desolating burnt landscape is the resulting scenario.

The interactive experience ends with the participant being presented with a screen showing information of the resulting damage in the style of a game score: as a measure of the damage generated in the *Play with Fire* interaction along with a snapshot and general data about how the performance fire damages, regeneration times and possible effects on climate change.

The participant picks up the phone from the deck where he/she had placed it when entering the room and is prompted to open a URL that contains a *Play with Fire* mobile web application that displays the damaged forest the participant leaves behind and its slow regeneration process. The duration of the forest regeneration and the length of time the application will accompany the participant will in fact depend on the data generated during the performance. Participants will carry it as a memento of their experience for a duration determined by the real-time interaction with the installation.

4. CONCLUSIONS & FUTURE WORK

Collaboratively developed by artists, activists and technologists, *Play with Fire* presents an innovative approach – with a challenging technological component – that comprises a strong dimension on social and natural sciences converging New Media Arts and Sustainability.

We have designed this interactive experience for persuasive purposes, inviting people to engage in forbidden and dangerous actions in a controlled environment, in order to confront them later with the long term consequences of their own choices.

After the conception, design and initial basic prototyping of *Play with Fire* [8], the main implementation of the interactive installation is currently underway and is scheduled to premiere in the Fall of 2011.

For the Creative Showcase and Interactive Art session at ACE 2011 we are presenting an interactive scaled down version of the installation that allows participants to *Play With Fire*

5. ACKNOWLEDGMENTS

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6. REFERENCES

- [1] *Ars Electronica Festival*. Linz http://www.aec.at/festival_history_en.php Retrieval date 2011/04/23.
- [2] Cipolla, C. and Peruccio, P. (ed.) 2008. *Changing the Change* – Conf. Proc., Torino, Allemandi, Conference Press.
- [3] *Games for Change* | Environment <http://www.gamesforchange.org/channels/environment> Retrieval date 2011/04/23.
- [4] La Molleindustria <http://www.molleindustria.org/en/home> Retrieval date 2011/04/23.
- [5] Madeira, R. 2009. *Parque*. Interactive installation, Parque da Paz – Almada, Portugal <http://vimeo.com/4789754> Retrieval date 2011/02/15.
- [6] Mendes, M. and Correia, N. 2010. *RTiVISS | Real-Time Video Interactive Systems for Sustainability*, MM'10 Proc. of the International Conference on Multimedia, ACM Multimedia 2010 – Interactive Art Program, Firenze, Italy.
- [7] O'Shea, C. 2009. *Hand from Above* <http://www.chrisoshea.org/hand-from-above> Retrieval date 2011/04/23.
- [8] *Play with Fire – A RTiVISS experience* | Work in progress <http://www.monicamendes.info/rtiviss/playwithfire> Retrieval date 2011/06/22.
- [9] Transmediale 2009. “Perish in Beauty? Climate Change as Cultural Demand”. *Annual Festival for Art and Digital Culture*, Berlin. <http://www.transmediale.de/perish-beauty-climate-change-cultural-demand> Retrieval date 2011/02/15.
- [10] Watson, T. and Gobeille, E. 2007. *Funky Forest*, Interactive Installation http://www.theowatson.com/site_docs/work.php?id=41 Retrieval date 2011/04/23.